Leakage-Safe Machine Learning for Water Quality Indexing

CCME Water Quality Index · 2.82M global records · AML Capstone, Phase 1

WQI: a 0–100 score of water health

Why WQI matters

Many lab tests needed at once → slow
One missing test → no score
Predict from routine data → fast score

GOAL

- Predict WQI from routine water chemistry
- Test on stations the model has never seen
- Handle the cap: 41% of scores sit at 100

1 Data
Clean rows, keep ambient water

→

2 Features
Oxygen deficit, N:P ratio, logs

→

3 Split
Site-held-out, zero overlap

→

4 Models
Ridge, Random Forest, XGBoost

→

5 Final
Test once, CI, SHAP, uncertainty

Stages 1–2 | Stages 3–4 | Stage 5 | Stage 6 | Stage 6 · results

Data and cleaning

- 1940–2023: 2.82M records, 5 countries, 61,689 sites
- Removed 9,971 duplicates; impossible readings set to blank
- Kept ambient water only: no effluent or sewage, 1990 onward
- **41%** of scores hit the cap of 100

1.99M clean records (from 2.82M raw)

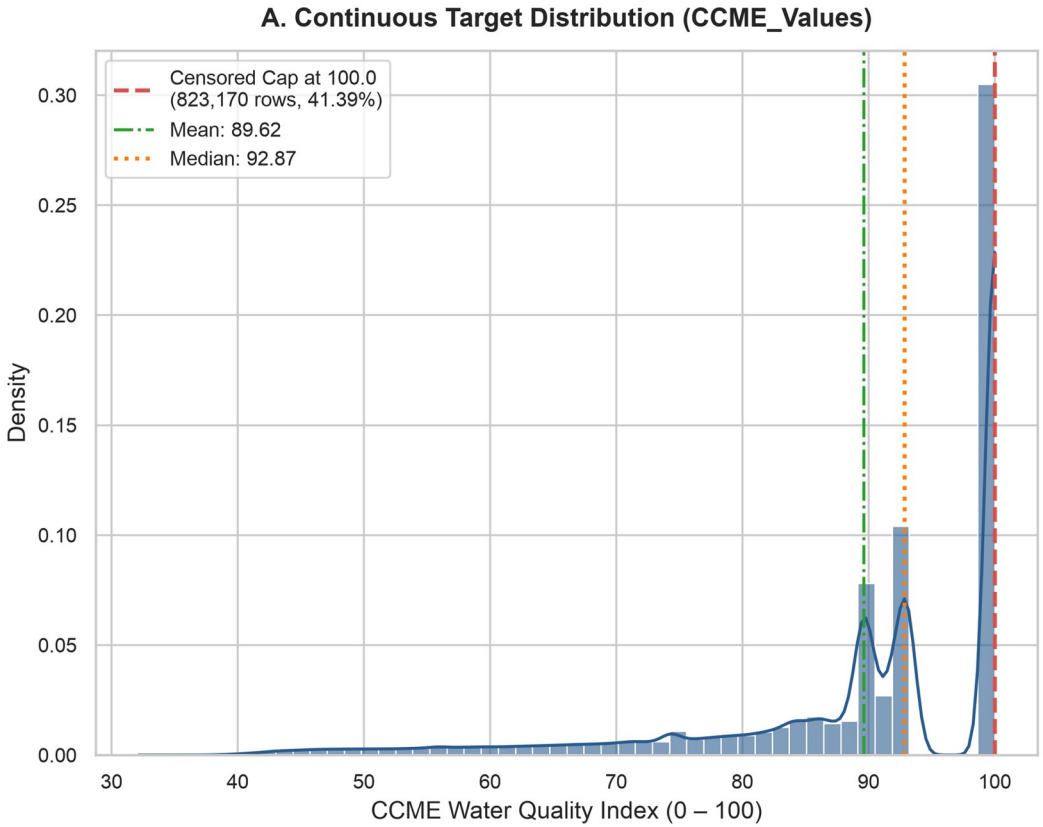


Figure 1A · WQI distribution: 41.39% of rows pile up at the cap of 100

Domain features

- Orthophosphate drives the score: rank correlation −0.913
- Oxygen is U-shaped: too low or too high both hurt
- 21 new features: oxygen deficit, N:P ratio, logs, month sin/cos
- **No leakage:** guideline-exceedance flags (the formula's own inputs) are kept out

0.92 Ridge R^2 with our features (was 0.52)

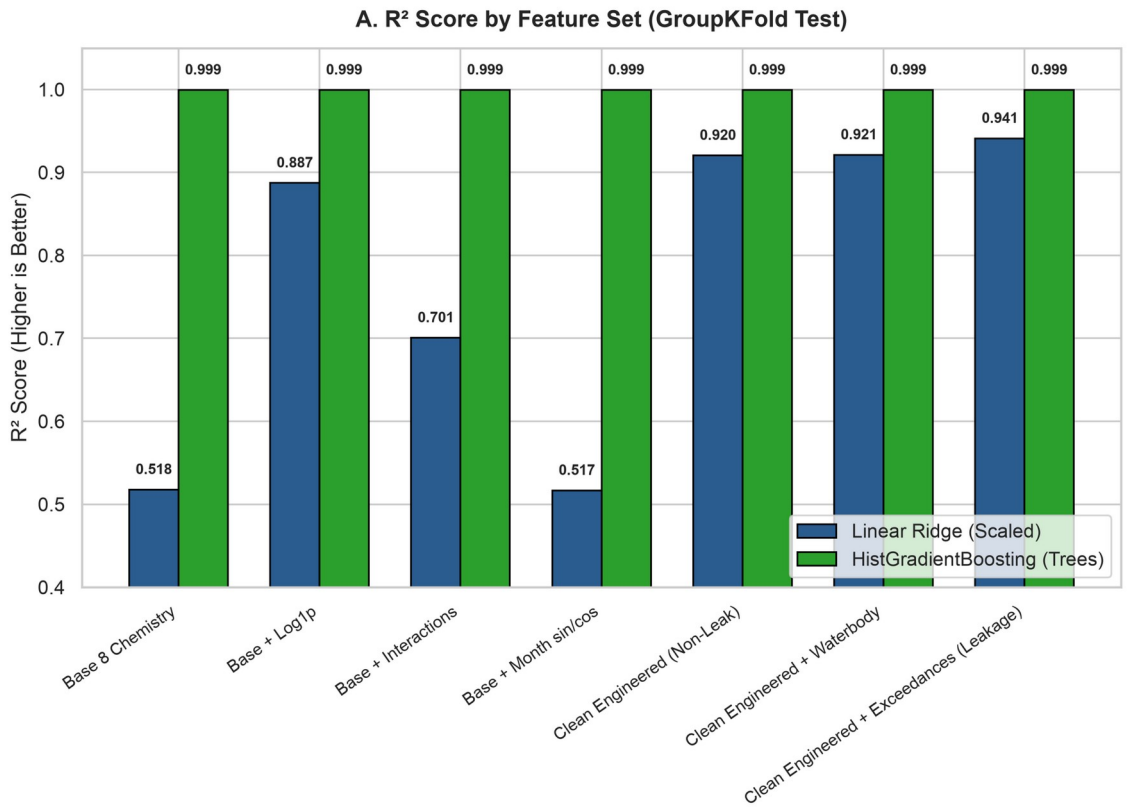


Figure 8A · R^2 by feature set: Ridge 0.52 → 0.92, trees stay at 0.999

Zero-leakage split

- **Random split**: 91% of test stations are also in training, so models memorize sites
- **Site-held-out**: split by station (group split on Area), so test sites are brand new
- **Proof**: 0 stations shared between train, validation and test
- **Also tested**: a time split, train 1990 to 2012, test 2018 to 2023

Split	Rows	Stations
Train	1,345,895	32,602
Validation	284,667	6,986
Test	358,067	6,987

Zero stations shared · checked by tests/test_splits.py

Comparing four models

- Ridge (linear): a straight-line baseline, test RMSE 3.68
- Random Forest: 100 trees averaged, test RMSE 0.36
- **XGBoost (clipped)**: lowest error of all four, clipped to 31.3–100
- **Hurdle**: two-stage, a near-tie but more complex

Check	Ridge	Random Forest	XGBoost (clipped)	Hurdle
Validation				
R^2	0.9234	0.9992	0.9995	0.9995
RMSE	3.7144	0.3802	0.2863	0.2921
MAE	2.6288	0.1245	0.0754	0.0781
Test				
R^2	0.9168	0.9992	0.9995	0.9995
RMSE	3.6841	0.3621	0.2834	0.2884
MAE	2.5855	0.1197	0.0704	0.0732

Final evaluation and what’s next

- **Best model**: clipped XGBoost, test RMSE 0.2834 (95% CI 0.268–0.299)
- **Across countries**: RMSE 0.17 Ireland, 0.21 USA, 0.41 England
- **Uncertainty**: the 80% interval covers 98.27% of true values
- **Phase 2**: physics-informed networks, TabNet, cross-country transfer

Check	Result	Verdict
Accuracy	R^2 0.9995, RMSE 0.28	Best of all models
Hard cases	MAE 0.64 when score < 45	Still accurate
Drivers	Phosphorus, ammonia, BOD	Real chemistry

Test set, used once · CI from 1,000 bootstrap repeats

0.28 WQI points: test RMSE (6,987 unseen stations)

Pipeline and code

Step	What it covers	Code
Data	Profiling, cleaning, ambient-water filter	cleaning.py, notebooks 01–02
Features	EDA, oxygen deficit, N:P ratio, ablation	eda.py, features.py, notebooks 03–04
Split	Site-held-out and time splits	splits.py, 05_splits_report
Models	Ridge, Random Forest, XGBoost, hurdle	models.py, notebook 05
Final	Bootstrap CI, quantile intervals, SHAP	models.py, 06_models_report

Code: github.com/varunudata/AML_Capstone